

Year	Commodity Contract	Traded Contract (in Lots)	SegmentName
2014	ALUMINUM MINI	368937	NON PRECIOUS METAL PRODUCTS
2015	ALUMINUM MINI	3554731	NON PRECIOUS METAL PRODUCTS
2016	ALUMINUM MINI	710599	NON PRECIOUS METAL PRODUCTS

Figure 1: Aluminium traded contracts (in lots) for 2014, 2015, 2016

```
sns.set_style("darkgrid") ..... line 6
bar_plot = sns.barplot(x=commodity["Year"],y=budget["Traded
Contracts (in Lots)"],
                      palette="muted",
                      order=commodity["Year"].tolist()) ..... line 7
plt.xticks(rotation=90) ..... line 8
plt.show() ..... line 9
```

We import the necessary libraries in Lines 1, 2 and 3. Here we are using Pandas, Seaborn and Matplotlib.

In Line 4, we declare a variable commodity and assign it the value read from the csv file. Pandas variable PD is used as the file handler to read the file. Next, in Line 5, we sort the values of the column 'Traded Contracts' in descending order. In Line 6, we set the style of the grid as 'darkgrid'. In Line 7 we declare a variable `bar_plot`, which will contain the values to be plotted on the x-axis and y-axis. We don't want a palette; so, we mute it and order it year-wise. We don't want our plot to be rotated; so we fix the rotation at 90. Finally, we need to show the bar plot in the window; hence, we call the `show` function.

Now run the program and you will get the image shown in Figure 2.

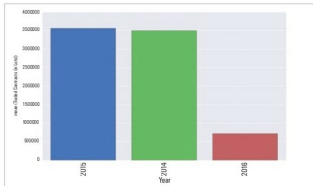


Figure 2: Histogram for traded aluminium contracts

1	Private flight	
2	Air Asia	
3	Sipho Jet	
4	Aeroflot Russian Airlines	
5	Qatar Airlines	
6	Lufthansa	
7	Air India	
8	Indigo Airlines	
9	Egypt Air	
10	Qantas	
11	Virgin Atlantic	
12	Fly Emirates	
13	Turkish Airlines	
14	Malaysia Air	
15	Austrian Airlines	
16	Army Air Corps	

Figure 3: Sample list of private airlines

This is one example of how we can generate bar charts easily.

Next, let's look at another example of a scatter plot visualisation in Python. For this example, we have sample data of the list of airlines in csv format. We will be plotting a scatter diagram to check the length of airlines' names against their IDs.

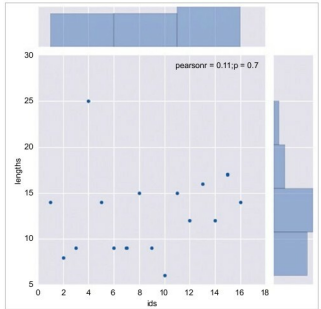


Figure 4: Scatter plot for ID vs private airlines based on length of name

Scatter_visualization.py

```
# Import the pandas library.
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Read in the airlines data.
airlines = pd.read_csv("airlines.csv", header=None, dtype=str)
# Provide Column Header names
airlines.columns = ["id", "name"]

name_lengths = airlines["name"].apply(lambda x: len(str(x)))
# plot scatter x axis as ids and y axis as length of names
data = pd.DataFrame({"lengths": name_lengths, "ids":
airlines["id"].astype(int)})
sns.jointplot(x="ids", y="lengths", data=data)
plt.show()
```

Once you run the above program, you will get the result shown in Figure 4.

Hint: There are many sites that offer different datasets in various formats. You can use https://catalog.data.gov/dataset?res_format=CSV for this purpose. It gives a list of various data in json, xml and csv format.

Python provides a lot of tools that are simple to use and can generate different kinds of data visualisation. Enjoy coding. 

By: Bhaskar Narayan Das

The author has over six years of experience in the IT industry and works as an application developer with IBM. He has worked with Java, Python and databases like NoSQL and other related technologies. You can contact him at bhaskarndas@gmail.com.